

SALEN-REUTENEN, Switzerland

WMO: 066230

Lat: 47.6511N

Lon: 9.0239E

Elev: 720

StdP: 92.97

Time Zone: 1.00 (EUC)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.5	-7.4	-10.2	1.7	-6.1	-8.6	2.0	-5.2	13.4	6.5	11.9	3.8	3.6	60	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	28.4	19.0	26.7	18.2	25.0	17.6	20.2	26.2	19.3	25.2	18.5	23.8	2.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	14.4	23.0	17.3	13.5	21.6	16.5	12.8	20.7	61.4	26.4	58.3	25.3	55.5	24.1	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.2	7.7	DB	-11.1	30.9	3.8	1.2	-13.8	31.8	-16.1	32.5	-18.2	33.1	-21.0	34.0	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	0.3	0.2	5.0	9.0	12.3	16.7	18.4	18.2	14.1	9.7	4.5	1.9		
	DBStd	7.61	3.91	5.00	4.03	4.53	3.97	3.86	3.67	3.60	3.26	3.92	3.87	3.48		
	HDD10.0	1309	299	275	160	73	23	1	0	0	3	54	169	253		
	HDD18.3	3443	558	508	413	281	191	76	44	49	132	266	414	511		
	CDD10.0	1036	0	0	5	43	93	203	262	253	126	46	6	0		
	CDD18.3	128	0	0	0	1	3	28	48	44	5	0	0	0		
	CDH23.3	776	0	0	0	3	14	191	299	259	11	0	0	0		
	CDH26.7	154	0	0	0	0	0	37	67	49	0	0	0	0		
(14) Wind	WSAvg	3.2	4.0	3.6	3.7	3.0	3.1	2.7	2.8	2.5	2.7	2.9	3.4	3.9		
(15) Precipitation	PrecAvg	971	56	54	61	68	106	114	113	102	83	72	71	72		
	PrecMax	1214	148	123	191	139	214	188	215	188	158	135	144	163		
	PrecMin	739	5	15	18	9	21	34	44	31	35	9	0	2		
	PrecStd	118	38	31	35	34	48	40	43	40	33	34	37	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	12.6	17.0	22.7	25.2	29.9	31.0	30.0	24.8	19.6	16.0	12.0		
		MCWB	7.2	6.5	9.4	13.4	16.4	19.8	19.9	19.4	16.7	13.5	10.2	6.6		
	2%	DB	8.6	9.8	15.1	20.1	22.6	27.7	28.5	28.0	23.0	17.9	13.8	9.6		
		MCWB	6.3	5.9	8.4	11.6	15.2	18.6	18.7	19.3	16.7	13.3	10.0	7.1		
	5%	DB	7.1	8.1	13.0	18.6	20.7	25.5	26.6	26.3	21.2	16.5	12.1	8.3		
		MCWB	5.1	5.1	7.5	11.3	14.1	17.9	17.8	18.7	16.1	12.9	9.5	6.6		
	10%	DB	5.3	6.5	11.2	16.4	18.9	23.5	24.6	24.4	19.4	15.1	10.3	7.1		
		MCWB	4.3	4.1	6.6	10.3	13.1	17.2	17.3	18.0	15.3	12.4	8.4	5.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	8.6	10.4	14.3	18.1	21.4	21.7	21.5	18.1	15.3	12.3	8.9		
		MCDB	9.3	10.1	15.4	21.0	22.6	27.7	28.2	26.3	22.7	17.7	14.7	9.5		
	2%	WB	6.8	6.7	9.1	12.7	16.6	19.8	20.0	20.2	17.2	14.4	10.7	7.7		
		MCDB	7.8	9.4	13.6	19.0	20.9	25.8	26.7	25.6	21.5	16.7	12.7	8.7		
	5%	WB	5.5	5.5	7.8	11.6	15.2	18.8	19.1	19.3	16.5	13.5	9.7	6.8		
		MCDB	6.6	8.0	11.4	17.3	18.8	24.3	24.8	24.9	20.2	15.6	11.5	8.5		
	10%	WB	4.5	4.4	7.0	10.8	13.9	17.7	18.2	18.5	15.6	12.7	8.6	5.9		
		MCDB	5.5	6.4	9.9	15.9	17.7	22.5	23.3	23.7	18.7	14.6	10.4	7.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.5	4.7	7.1	7.9	7.9	8.1	8.1	8.0	6.9	5.5	4.0	3.7		
		MCDBR	5.4	7.7	10.6	10.8	10.6	10.5	10.4	10.5	8.7	7.1	6.5	4.9		
	5% WB	MCWBR	4.1	4.7	5.0	4.9	4.9	5.2	4.1	4.3	4.0	4.3	4.0	3.3		
		MCWBR	5.0	7.0	8.1	9.4	9.4	10.1	9.3	9.3	7.8	6.4	5.9	4.3		
(40) Clear-Sky Solar Irradiance	taub	taub	0.282	0.304	0.336	0.369	0.381	0.386	0.388	0.382	0.358	0.332	0.303	0.277		
		taud	2.467	2.408	2.386	2.323	2.324	2.342	2.328	2.363	2.413	2.477	2.508	2.471		
	Ebn at Noon	Ebn at Noon	813	862	883	882	882	877	869	860	849	819	778	772		
		Edh at Noon	64	84	100	118	122	121	121	112	97	78	62	56		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.11	1.93	3.14	4.31	5.00	5.73	5.50	4.81	3.59	2.24	1.23	0.91		
	RadStd	RadStd	0.12	0.27	0.42	0.61	0.51	0.52	0.50	0.49	0.37	0.24	0.15	0.14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (112 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.69	+0.43	+0.36	N/A	N/A	+55	+14

Nomenclature: See separate page